

Classification of XAI Techniques

Category	Types of Techniques	Examples
By Timing	- Ante-hoc (built-in), Post-hoc (after training)	Decision Trees, SHAP, LIME
By Scope	- Global (whole model), Local (specific prediction)	Feature Importance, LIME, SHAP
By Model Dependency	- Model-agnostic, Model-specific	SHAP (agnostic), Grad-CAM (specific)
By Representation	- Visual , Textual, Numerical	Saliency maps, rules

Post-hoc XAI Techniques

Technique	Description
LIME	Explains individual predictions using a simple local surrogate model
SHAP	Provides feature contribution values using Shapley value theory
Grad-CAM	Visualizes image regions important for CNN decisions
Saliency Maps	Highlights important pixels for image classification
Counterfactuals	Explains by showing minimal changes needed to alter the prediction
Feature Importance	Ranks global importance of input features
Decision Rules	Converts model behavior into if-then logic

Local Interpretable Model-Agnostic Explanations (LIME)

- A LIME (Local Interpretable Model-Agnostic Explanations) is an XAI technique that explains individual predictions of any machine learning model — locally, using a simple interpretable model (like a linear model or decision tree) as a surrogate.
- LIME (Local Interpretable Model-Agnostic Explanations) is a method for explaining the predictions of machine learning models by approximating them locally with interpretable models
- It provides users and developers with insightful, human-readable reasons behind AI decisions — making it one of the most widely used tools in trustworthy AI.
- LIME helps make black-box predictions explainable for a specific instance by fitting an interpretable model locally.

What is LIME?



Local

Local neighborhood of the instance

Interpretable

A human should be able to interpret

Model-agnostic

Applicable for all models

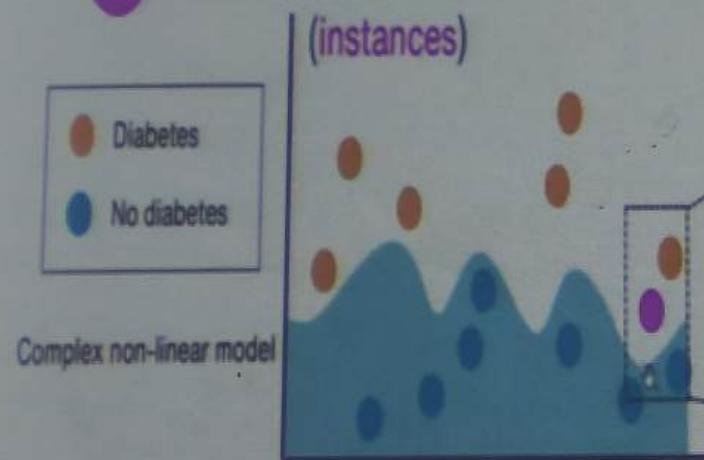
Explanations

Explanation that helps the interpretation

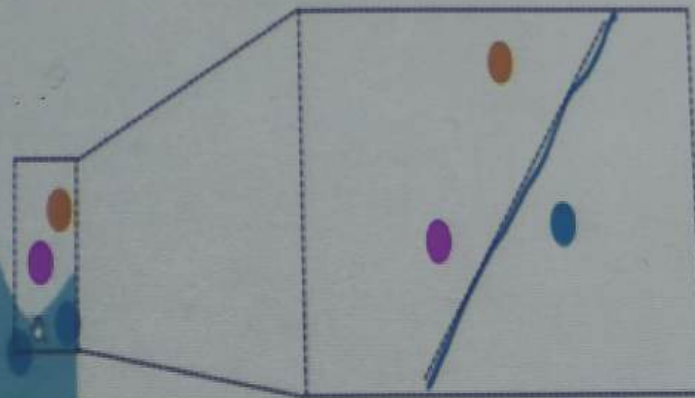
LIME- Steps:

1. Pick an instance
2. Create synthetic data by slightly modifying the input (perturbations)
3. Use the original black-box model to make predictions on this new data
4. Train a simple surrogate model (e.g., linear) on this dataset
5. Interpret the surrogate model to understand what influenced the original prediction

1 Global model, local data point

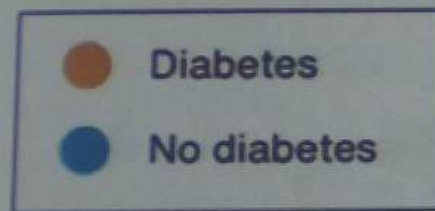


2 Local linear model near the point

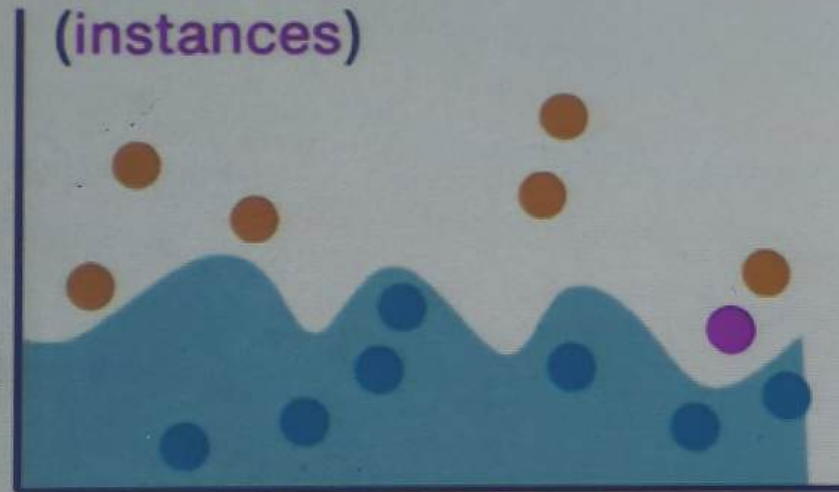


1

Global model, local data point
(instances)



Complex non-linear model

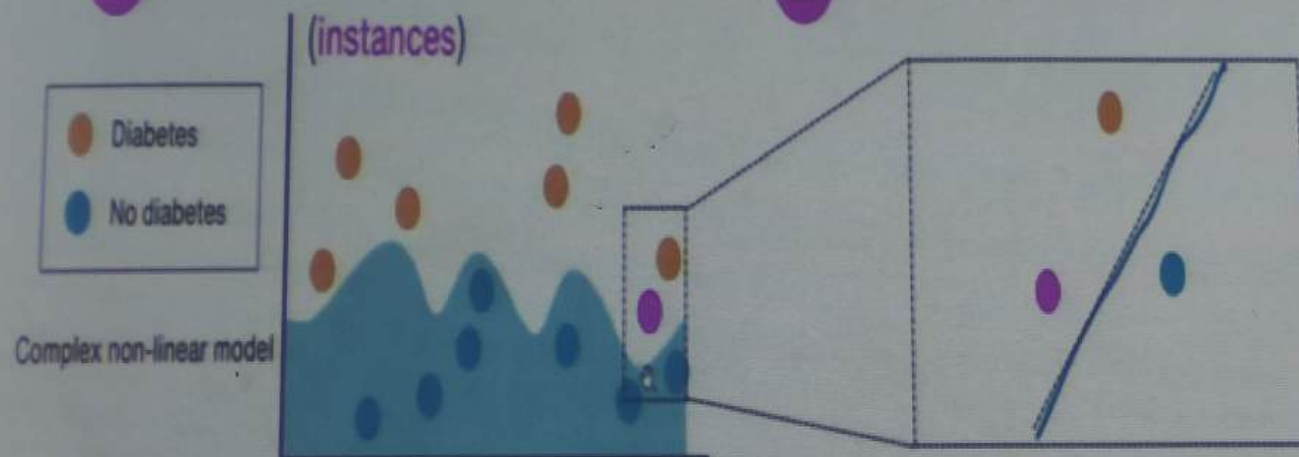


1

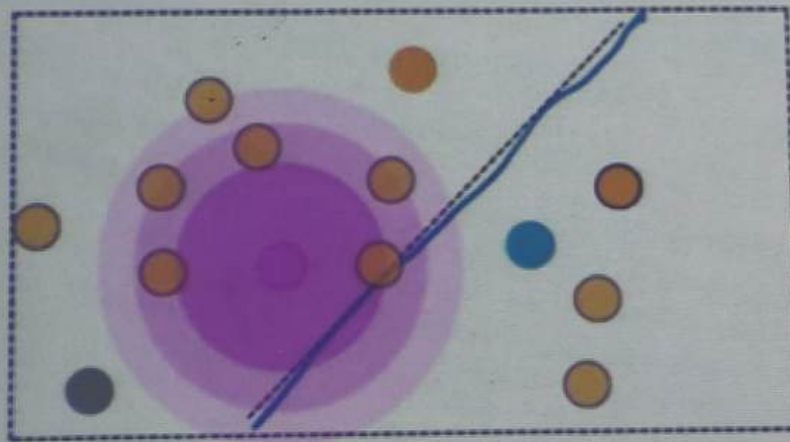
Global model, local data point

2

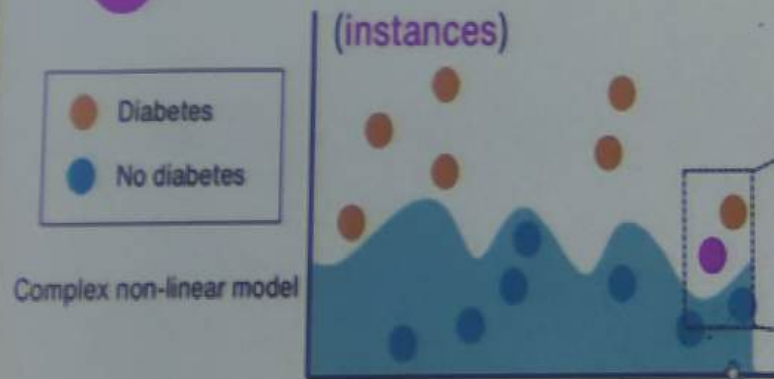
Local linear model near the point



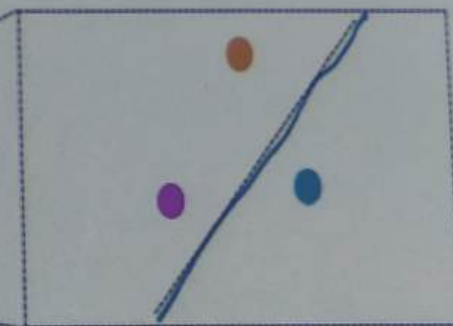
3 Perturbed data weighted according to the distance to the local point



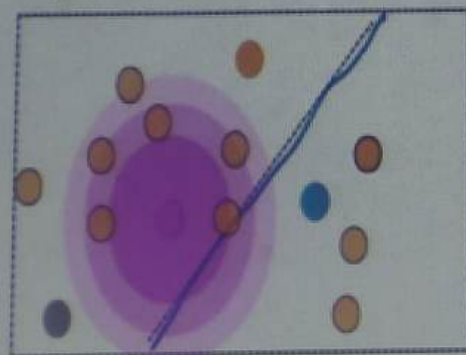
1 Global model, local data point



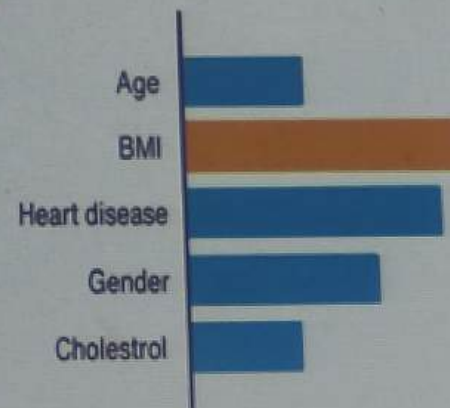
2 Local linear model near the point



3 Perturbed data weighted according to the distance to the local point



4 Features relevant to the data point

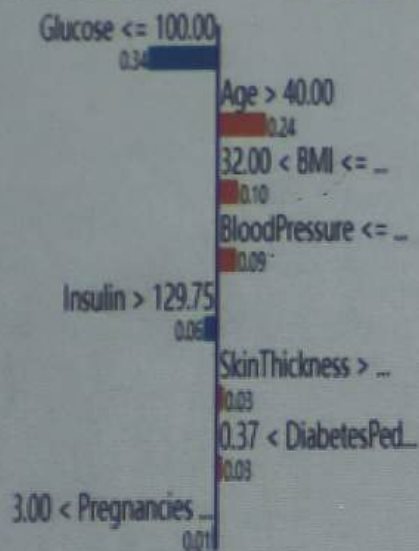


Prediction probabilities



No Diabetes

Diabetes



Feature Value

Glucose	98.00
Age	43.00
BMI	34.00
BloodPressure	58.00
Insulin	190.00
SkinThickness	33.00
DiabetesPedigreeFunction	0.43
Pregnancies	6.00

LIME

Example: Email Spam Classification Using LIME

You've trained a complex black-box model (e.g., Random Forest or XGBoost) to classify emails as "Spam" or "Not Spam" based on features like:

- Presence of certain words (free, money, win, etc.)
- Length of email
- Number of exclamations (!)
- Number of capital letters
- Time of day it was sent

The model performs very well — but you don't know *why* it labels a specific email as "Spam".

Let's say you input the following email:

"Congratulations! You've won ₹1,00,000! Claim your prize now."

And your model predicts: **Spam**

Now you apply **LIME** to explain this prediction.

LIME

Example: Email Spam Classification Using LIME

Steps:

1. Perturb the Email Text

LIME generates **slightly altered versions** of the email by:

Removing some words (e.g., delete "₹1,00,000")

Replacing or masking "Congratulations", "prize", etc.

Keeping structure but varying features

2. Get Model Predictions

For each modified version, it asks the original model:

Is this spam or not?

Get probability score (e.g., 85% spam)

LIME

Example: Email Spam Classification Using LIME

Steps:

3. Fit a Simple Surrogate Model

LIME trains a logistic regression model on this new dataset — it's local and only cares about the vicinity of the original email.

4. Interpret the Surrogate Model

LIME shows which words contributed most:

Word	Contribution
"won"	+0.30
"₹1,00,000"	+0.28
"Congratulations"	+0.25
"prize"	+0.20
"claim"	+0.15
(others)	Near zero

LIME

Essentially, LIME creates a simplified, local view of the complex model's behavior, making it easier to understand why it made a particular prediction for a specific instance. This helps build trust in AI systems by making their decisions more transparent and understandable.

LIME

Advantage(s)

Flexible

Easy to implement

Human-understandable

Useful for debugging

Why It's Useful

Works with any model or data type (tabular, image, text)

Widely supported in Python libraries (e.g., lime package)

Uses simple local explanations (easy to interpret)

Developers can understand why models behave unexpectedly

Limitation(s)

Local approximation only

Unstable explanations

Computational cost

Description

Doesn't give a full global understanding of the model

Results may change with different perturbations or sampling

Requires many perturbations and predictions

SHAP (Shapley Additive Explanations)

- SHAP (SHapley Additive exPlanations) was introduced by Scott Lundberg and Su-In Lee In the paper: "A Unified Approach to Interpreting Model Predictions" published: 2017 at NIPS (NeurIPS) conference.
- Link: <https://arxiv.org/abs/1705.07874>
- SHAP is a method for explaining predictions using additive combinations of Shapley values.
- SHAP is based on game and introduces to unify, improve, and fairly explain any ML model.
- Model Agnostic: It can be applicable to any machine learning model.
- SHAP can provide both local explanations (explaining a single prediction) and global explanations (understanding overall feature importance and relationships across the entire dataset).

SHAP- Game Theory Overview

Game Theory is a branch of mathematics that studies how players make decisions in competitive or cooperative settings.

It's used to analyze situations where:

- Multiple players are involved
- Each player's outcome depends on their own action and others' actions

Key Concepts:

- Players: The decision-makers (e.g., companies, people, features)
- Coalitions: Groups of players working together
- Payoff: The reward a player (or group) receives
- Shapley Value: A fair way to divide total payoff among players, based on their contribution

SHAP- Game Theory Overview

Ex. Imagine your college cricket team won a match and ₹10,000 as a team reward. But how do you fairly distribute it?

- Player A (batsman) scored 60 runs
- Player B (bowler) took 3 wickets
- Player C (fielder) saved 20 runs

You need a fair formula to assign credit or reward to each player. That's exactly what Shapley values do and SHAP uses this idea.

How Game Theory Relates to SHAP?

Game Theory	SHAP Explanation
Players	Features (e.g., income, age, credit)
Game	Making a prediction using a model
Payoff	The output of the model (e.g., 34%)
Contribution of a player	Feature's impact on the final prediction
Shapley Value	Fair share of contribution from a feature

SHAP-Mathematically

$$f(x) = \phi_0 + \sum_{i=1}^n \phi_i$$

Where:

- $f(x)$: final prediction
- ϕ_0 : average prediction (baseline)
- ϕ_i : **SHAP value** for feature i

SHAP

Feature	Contribution
Base value	50%
Age = 45	+6%
Income = ₹80k	+12%
Loan amount = ₹10L	-8%
Credit = Bad	-26%
Final Prediction	34%

SHAP tells you what increased and what decreased the prediction from 50% base to 34%.

SHAP

Ex. Model task: Predict if a person will repay a loan. Suppose your model says: "This person has a 34% chance of repaying the loan."

Now SHAP breaks it down like this:

Feature	Contribution
Base value	50%
Age = 45	+6%
Income = ₹80k	+12%
Loan amount = ₹10L	-8%
Credit = Bad	-26%
Final Prediction	34%

SHAP tells you what increased and what decreased the prediction from 50% base to 34%.

SHAP- Applications:

- Model Debugging: Identifying biases or unexpected behavior in a model.
- Feature Importance Analysis: Understanding which features are most influential in a model's predictions.
- Individual Prediction Explanation: Providing clear reasons for specific predictions to end-users or stakeholders.
- Model Comparison: Comparing the feature contributions across different models.

Why XAI in Healthcare?

- Medical decisions (like cancer detection, heart disease risk) are life-critical.
- Doctors and patients need to trust AI outputs.
- Black-box models (like deep CNNs) may be accurate but not acceptable without explanations.

Example:

- AI predicts a tumor is malignant (cancerous).
- Using Grad-CAM, doctors can see that the model focused on the tumor region in MRI → builds confidence.
- Without explanation, doctors may reject the AI system.

Why XAI in Healthcare?

How XAI Helps:

- Highlighting key features (e.g., specific symptoms, MRI pixels)
- Counterfactuals: “If the tumor size were smaller, prediction changes to benign.”
- Transparency for physicians → reduces liability and improves adoption.

Making AI Decisions Trustworthy for Physicians and Patients

- Physicians: Want scientific reasoning behind AI recommendations.
- Patients: Need human-understandable language explanations (layman terms).

Examples:

- Physician: AI diagnosis = "diabetes" then explanation should mention blood sugar levels, BMI, family history.
- Patient: Same explanation can be simplified as "Your blood sugar is high, which increases diabetes risk."

Making AI Decisions Trustworthy for Physicians and Patients

- XAI Provides multi-level explanations: Technical (for doctors), Simplified (for patients)
- It increases trust in adopting AI-based decision support systems.
- It ensures accountability (doctors can defend AI-based diagnoses in case of disputes).

Sales Predictions on the House Sale

- AI models predict house prices based on features like location, size, age, amenities.
- Buyers, sellers, and banks need to understand why a certain price is suggested.

Examples:

Black-box model predicts: "House Price = ₹75,00,000"

SHAP Explanation:

Location (+₹20L)

Size (+₹15L)

Old age (−₹5L)

No parking (−₹2L)

Buyers/sellers trust prediction since feature contributions are transparent.

Sales Predictions on the House Sale

XAI Benefits:

- Increases customer confidence.
- Banks can justify loan approvals.
- Prevents bias (e.g., model unfairly lowering prices in certain regions).

Recent Trends in Trustworthy XAI

Human-Centric Explanations

Focus on *how people* interpret explanations, not just math.
Example: Using natural language explanations instead of graphs.

Causality-driven Models

Moving beyond correlations → focus on cause-effect relationships.

More robust and fair.

Hybrid Models

Combining deep learning + rule-based reasoning for both accuracy & transparency.

Recent Trends in Trustworthy XAI

Interactive XAI

Human-in-the-loop systems where doctors, financial experts, or customers can adjust or query the model.

Regulatory Push

GDPR (Europe) & AI Act → enforce “Right to Explanation” for users affected by AI decisions.